

MACHINE LEARNING FROM SCRATCH

POLYNOMIAL REGRESSION



The Most Confusing Data Science Interview Question

Why is Polynomial Regression still called Linear Regression?

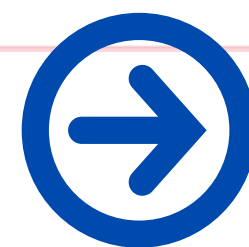
Let's simplify it.

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THE PROBLEM



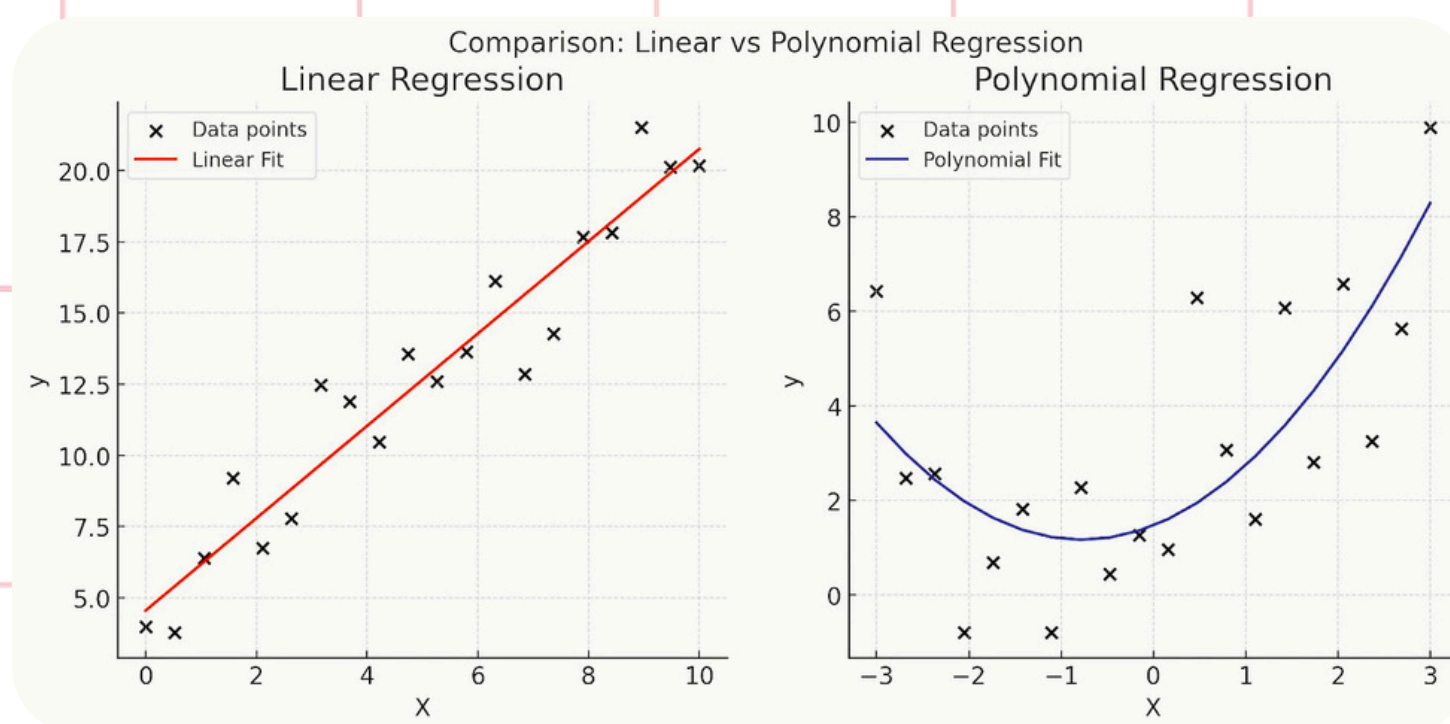
Linear Regression works well when the relationship between variables is a **straight line.**

But many **real-world datasets don't behave that way.**

Think about:

- **Salary vs Experience**
- **Temperature vs Ice Cream Sales**
- **House Prices**

Many relationships are **curved rather than **Perfectly linear**.**



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What is Polynomial Regression?



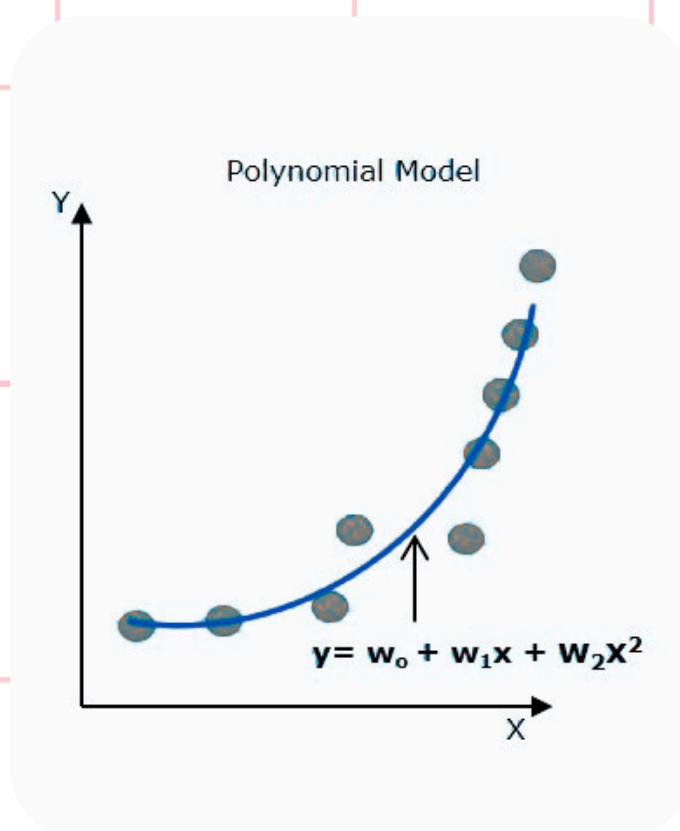
Polynomial Regression extends Linear Regression by creating new features from the original feature.

Instead of using only x :

It also creates:

- ✓ x^2
- ✓ x^3
- ✓ x^4

This allows the model to fit curved relationships.

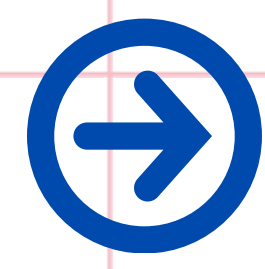


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The Biggest Misconception



Interview Question

Why is Polynomial Regression still considered Linear Regression?

Because the model is linear in its coefficients, even though the input features are transformed.

$$y = B_0 + B_1x + B_2x^2$$

- **The Exponents (x^2):** make the plot curved.
- **The Coefficients (B):** are what the model solves for. Because they are all to the power of 1 (strictly linear), the underlying mathematics remains Linear Regression.

That's why the underlying algorithm is still Linear Regression.



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Choosing the Degree

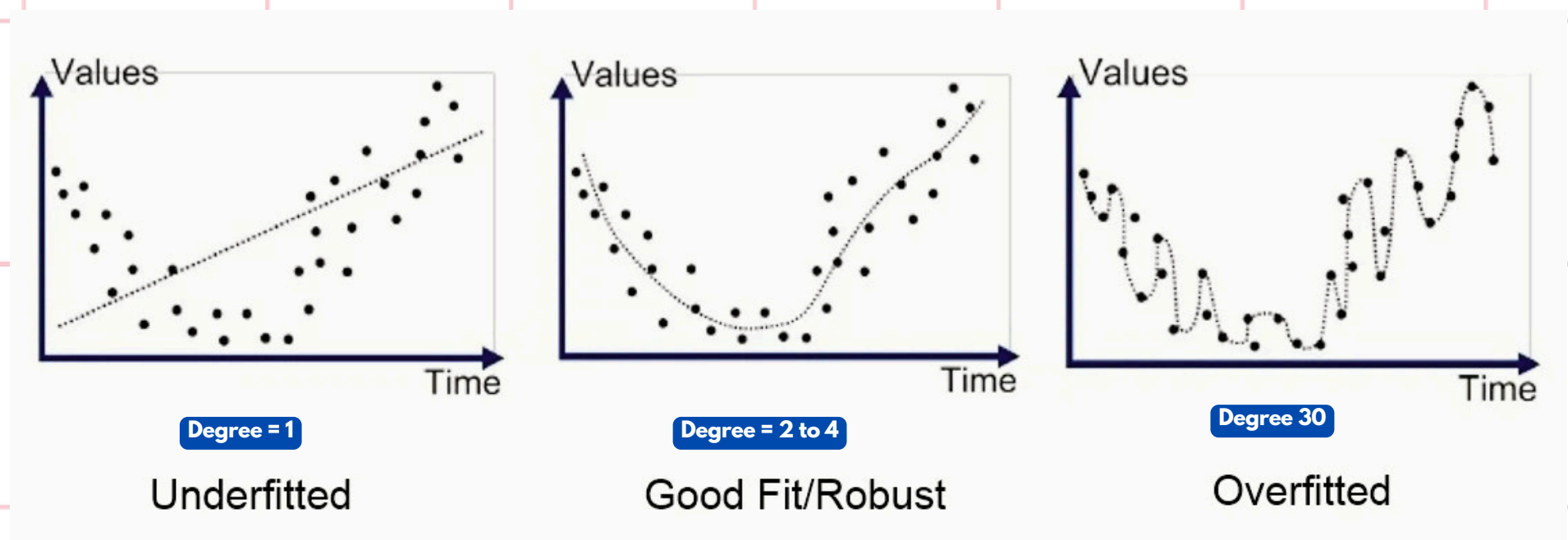


Degree 1 → Underfitting

Degree 2 or 4 → Usually a good balance

Very high degree → Overfitting

Higher complexity doesn't always mean better performance.



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My Practice



To strengthen my understanding, I implemented Polynomial Regression on:

- ✓ **A custom dataset**
- ✓ **Temperature vs Ice Cream dataset**

using Python and scikit-learn.

Comparing Linear Regression with Polynomial Regression helped me understand when a curved model performs better.

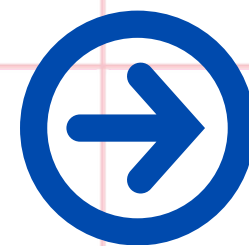


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Key Takeaways



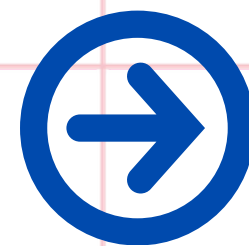
- ✓ **Polynomial Regression creates new polynomial features.**
- ✓ **It captures nonlinear relationships.**
- ✓ **It remains linear in its coefficients.**
- ✓ **Choosing the right degree is important to avoid overfitting.**



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What's Next?



Next in this series:

Bias-Variance Trade-off

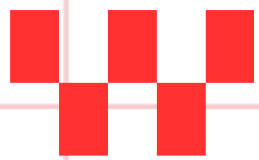
- **How do we know when a model is too simple or too complex?**

Follow along as I continue learning Machine Learning from scratch.

GitHub implementation linked in the comments.

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Thanks for Reading!



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